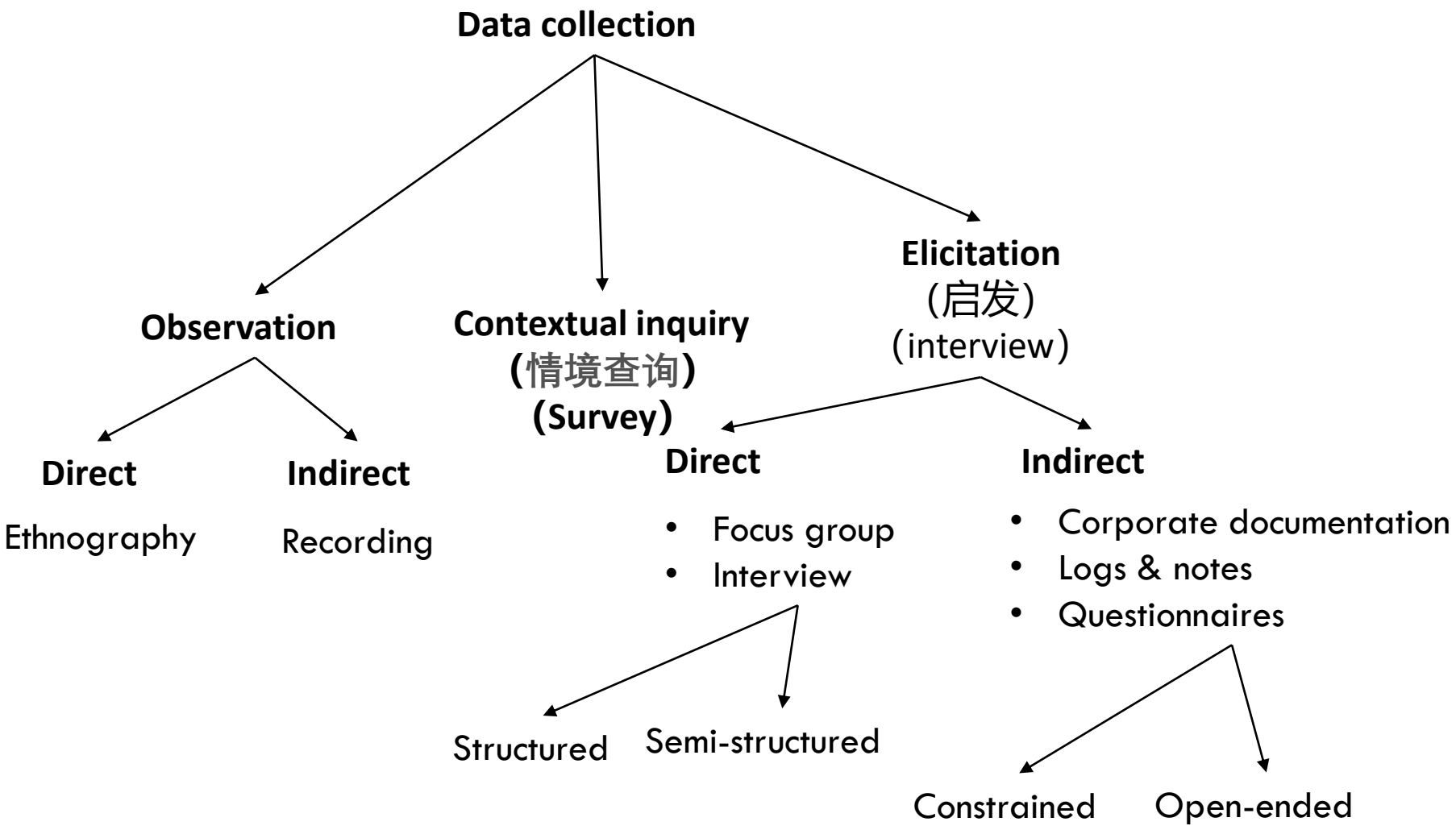
Human-Computer Interaction

6. Mental Model (1)



Recap: Data collection methods



Recap. Interview



A method of asking questions & listening

- Ask what you can't observe
- Hear about people's thoughts and experiences:
  - What they do
  - How they do things
  - Their opinions on current activities
  - How much they like one thing compared with another

Recap: Pros & Cons of Interview

- Pros
  - "Deep" rich data
  - Interactive
  - Open-ended
  - Exploratory
  - More economical in some cases than observing the user
- Cons
  - Time-consuming (both in conducting interviews and doing analyses)
  - Can be inconsistent and subjective
  - Amount of data can overwhelm
  - Interviewee can be biased by interviewer, or what they say might be different from what they do

Recap: Interview Questions

1. Get specific
   - Ask for tasks, roles and details of tasks
   - Ask about specific moment in the past: walk through tasks "yesterday"
   - Ask for examples!!!
   - Example: "What goes through your head when an online purchase fails?" vs. "Tell me what went through your head the last time you tried to buy something online and the purchase failed."
2. Don't ask questions asking frequency (How often you do X?)
3. Don't ask how much they like things on an absolute scale.
4. Don't ask questions which you can easily predict the answers
5. Don't ask questions that will influence the answer
   - Example: "How anxious do you feel when an online purchase can't be completed successfully?" vs. "Try to remember the last time an online purchase couldn't be completed for some reason. How did you feel then?"
6. Don't ask what they would do/like/want; new features

Recap. Structure of Interview Script

1) Introduction

   Example: "Hi Alice, thanks for coming down. My name is _____, and I'll be conducting the interview today. We're doing some market research, so some of the questions might be a little personal, and will concern your lifestyle. If you feel uncomfortable answering any of the questions, just let me know and we can skip it. I can't tell you much about what we do or the exact reason we're doing this research now, because it might affect the way you answer the questions. But I can definitely tell you after the interview is over. Do you have any questions so far? If you don't mind, I would record the audio for this interview; it will only be used internally, for the purposes of this research."

2) Warm up questions
   - Examples of warm up questions: "What's your occupation / what do you do all day?", "Can you tell me more about your hobbies?", "How often do you use the internet? etc."

3) Main body questions
   - Start with questions relating to specific incidents in the past, before moving on to speculative ones.
   - Follow a flow that makes sense to the respondents

4) Wrap up
   - Give your respondent a sense of closure.
   - Finally, thank them for their time and contribution towards your research.

Recap: Interview: Ask More!

- Ask "Why?" and "How?" until you get an interesting answer or subject gets frustrated.
- "Tell me more!": Ask for more details on anything that sounds interesting
- Repeat/rephrase/summarize to make sure you understood

Follow-up questions

- Ask why
- Ask for an example
- Clarify terms
- Ask for step-by-step actions
- Make the interview to be about them, not you

Recap: Survey

Types of survey questions

- Mutually exclusive choices (radio buttons)
- Non-mutually exclusive choice (checkboxes)
- Ranges (overlapping, open-ended)
- Likert Scale
- Short answer fill-ins
- Comments

Question Type	Example
Choose one	Which of the following sports is your favorite? ○ Baseball ○ Basketball ○ Football ○ Hockey ○ Other _____ ○ None
Choose many	Which of the following sports do you play regularly? ☐ Baseball ☐ Basketball ☐ Football ☐ Hockey ☐ Other _____ ☐ None
Rating	Please rate your overall level of enjoyment watching your favorite sport on television. Completely Dissatisfied ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ Completely Satisfied 1 2 3 4 5 6 7 8 9 10
Ranking	Please rank these sports in order from your favorite (#1) to least favorite (#4). ___ Baseball ___ Basketball ___ Football ___ Hockey
Open-ended	Please describe why [xxx] is your favorite sport? _____ _____ _____

Recap: Likert Scale

- Best used to measure and evaluate customer sentiment on a specific product, service or experience
- Response labeling is important!
- Common Likert Scale questions:
    1. Agreement
    2. Likelihood
    3. Satisfaction
    4. Importance

Recap: Interview or Survey?

Interviews

- Explore a domain
- Probe & follow-up
- Capture multiple data sources
- Rapport-building → Can ask about sensitive issues

But:

- Expensive
- Requires coding to interpret data
- Difficult to make comparisons

Surveys

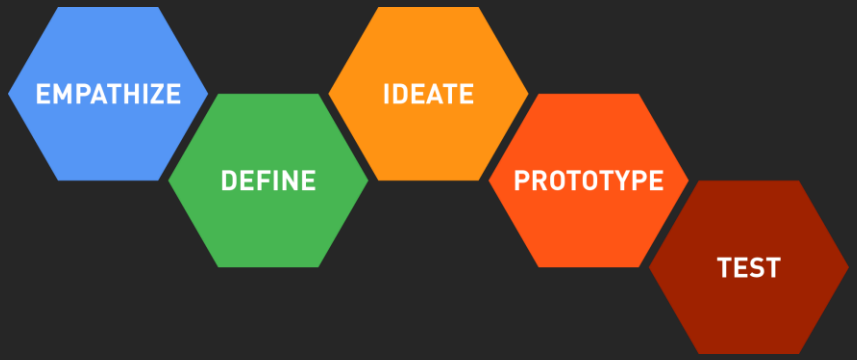
- Defined sample from larger population → Generalizable
- Comparatively low cost
- Ask multiple questions efficiently
- Standardized data allows comparisons
- Quantitative results

But poor for:

- Starting research
- Determining questions to ask
- Clarifying questions for respondent
- Following up unexpected comments

Today's agenda

Mental models



Mental models（心智模式）

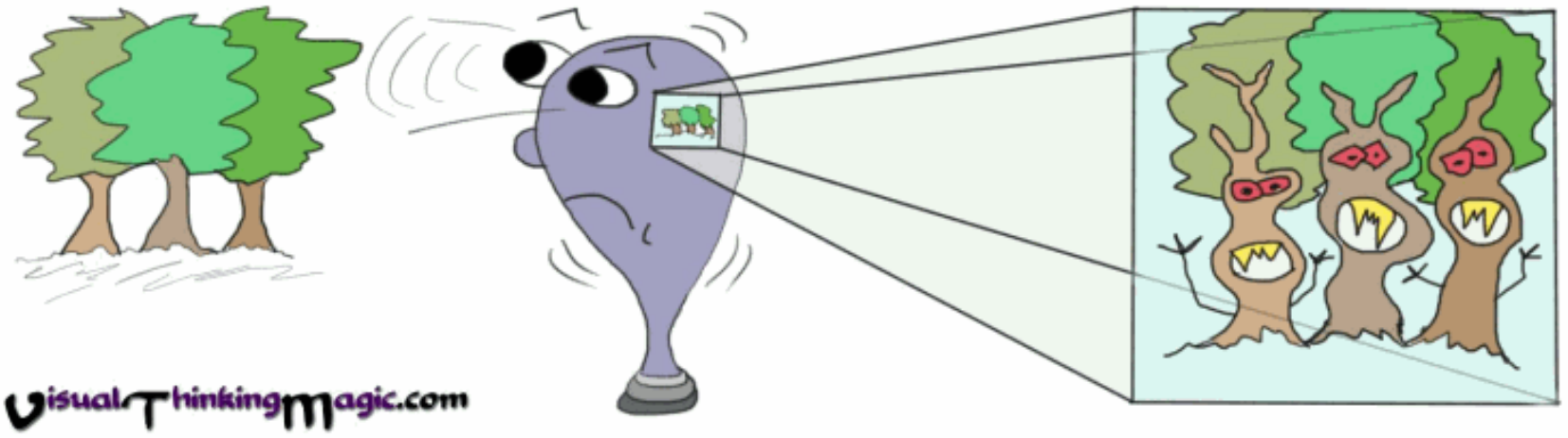
Obesity is due to eating too much and not enough exercise. The patient is to blame, so solutions must start with the patient. The biggest concern is diabetes.

I'm fat because my metabolism is slow, hormone and gland problems and too much stress. Health professionals should help me lose weight. I am most worried about difficulties getting to work.



Mental model

"How your mind thinks that something works"



You form mental models about everything you interact with.











Mental models are abstract representations.









Mental models let you make predictions.











Mental model

"The models people have of themselves, others, the environment and the things with which they interact. People form mental models through experience, training and instruction"

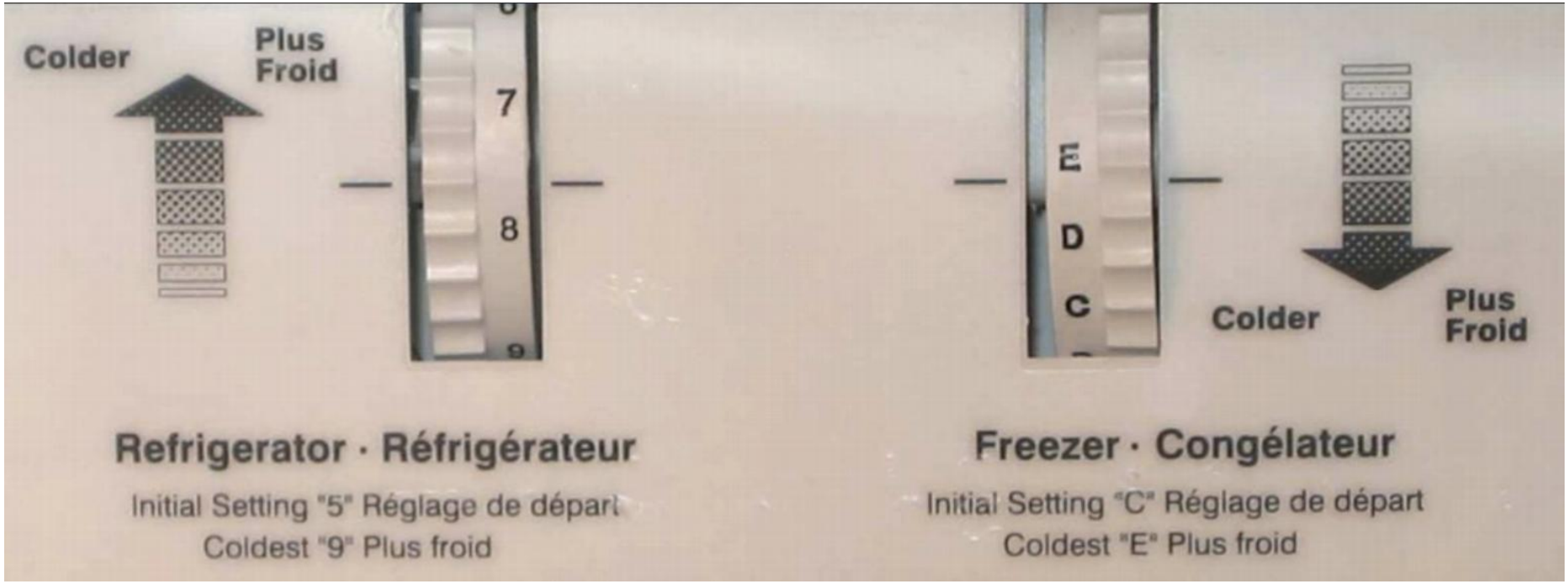
- An explanation of someone's thought process about how something works in the real world
- Enable people to reason about a system
- Affect the way we see and interpret reality
- When users approach an unfamiliar system, they subconsciously refer to their mental model, User's model

"To break a mental model is harder than splitting the atom."

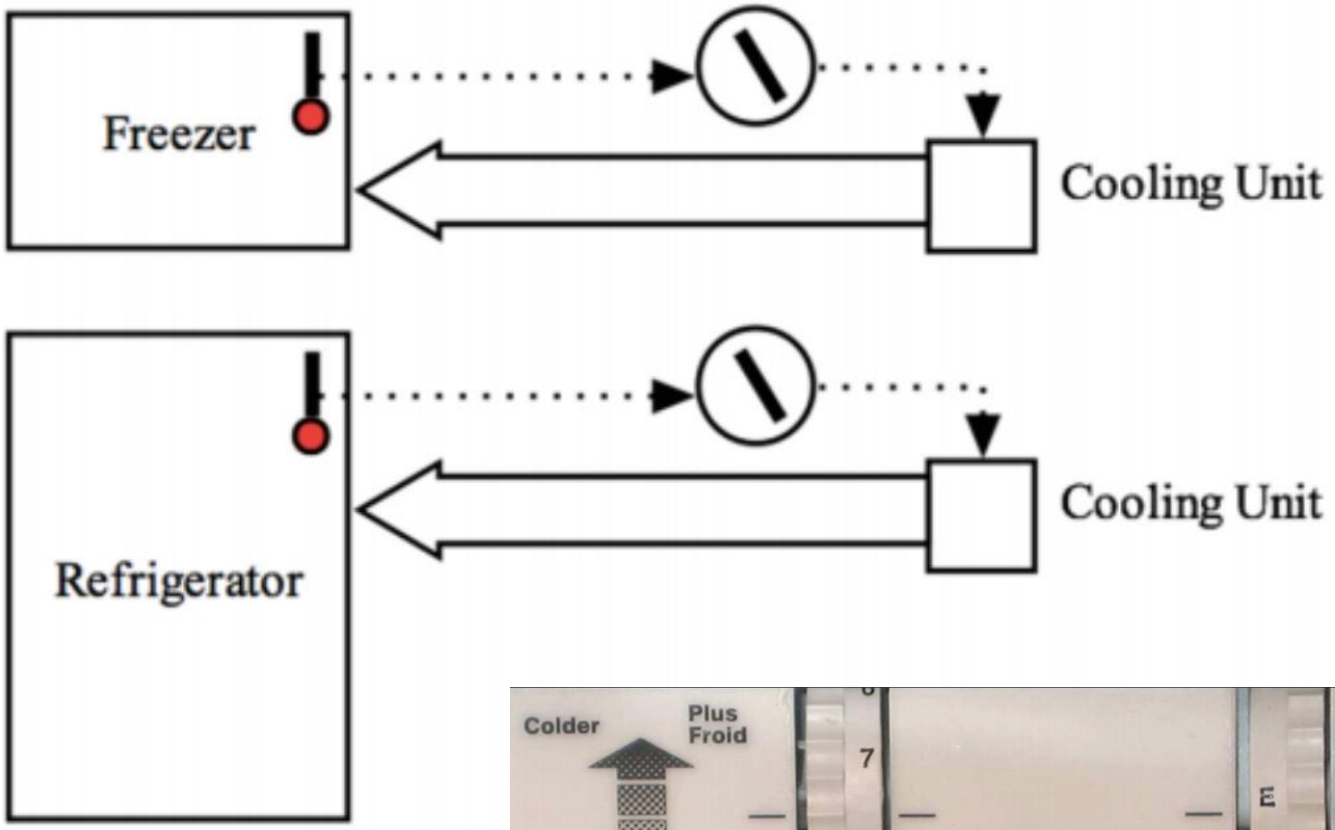
--Albert Einstein

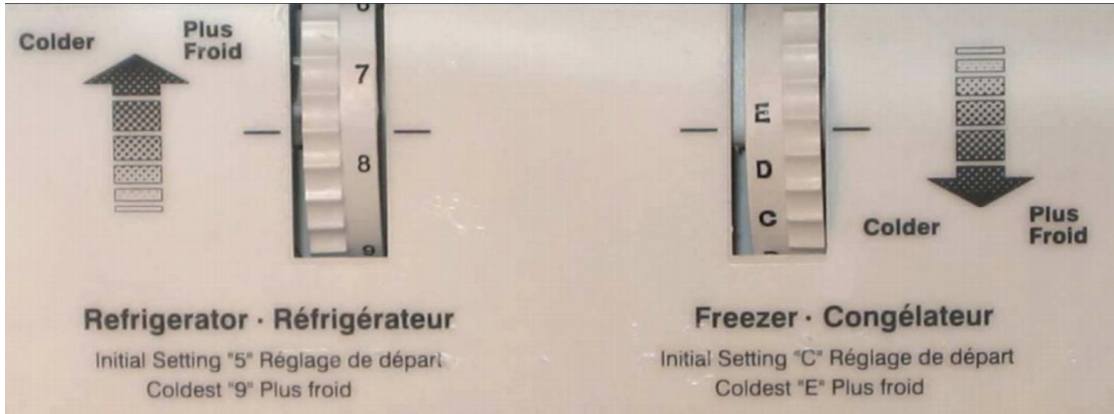
Let’s make some predictions

What will happen to the refrigerator when you turn the left dial up to “colder”?
What will happen to the freezer?

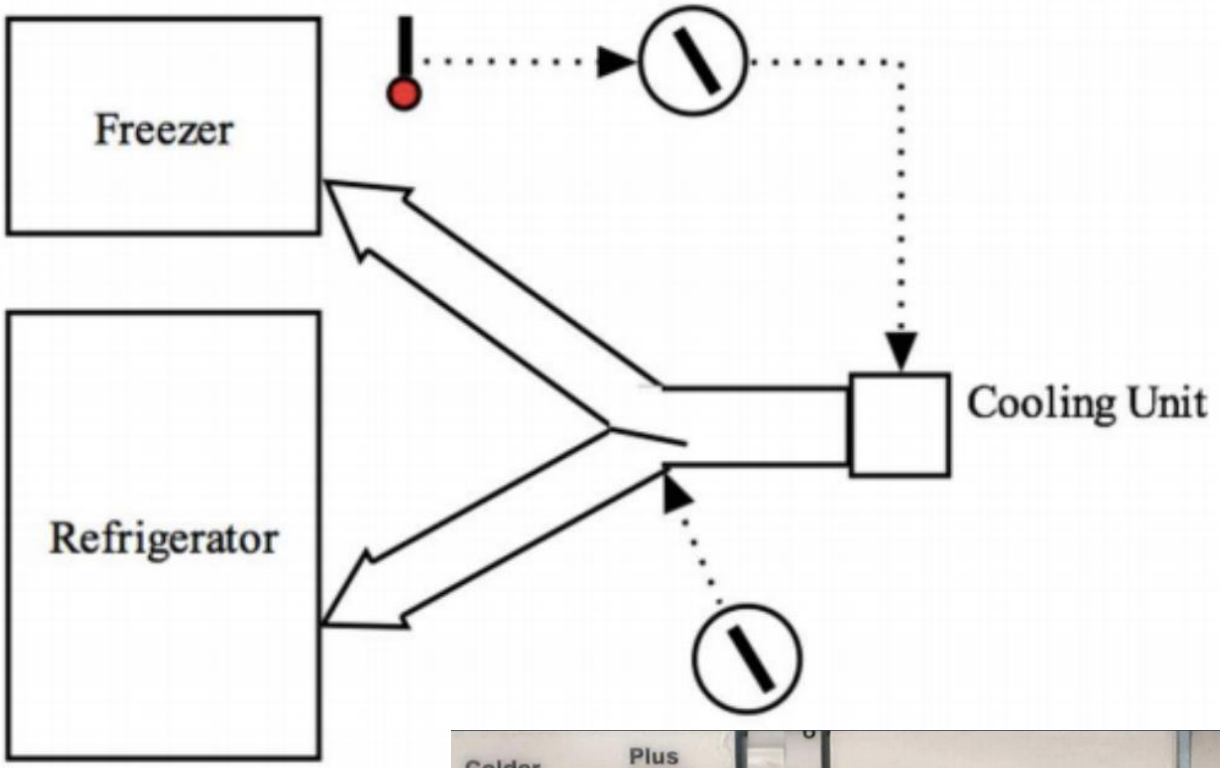


One reasonable mental model:

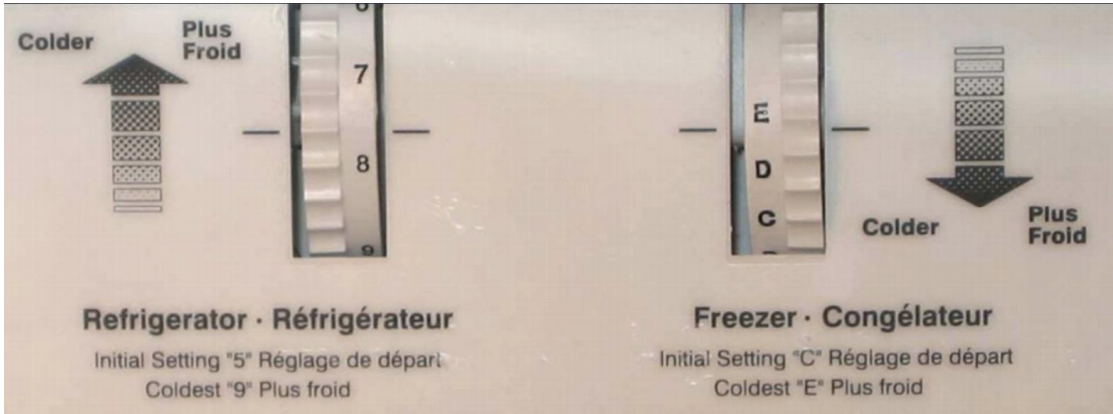




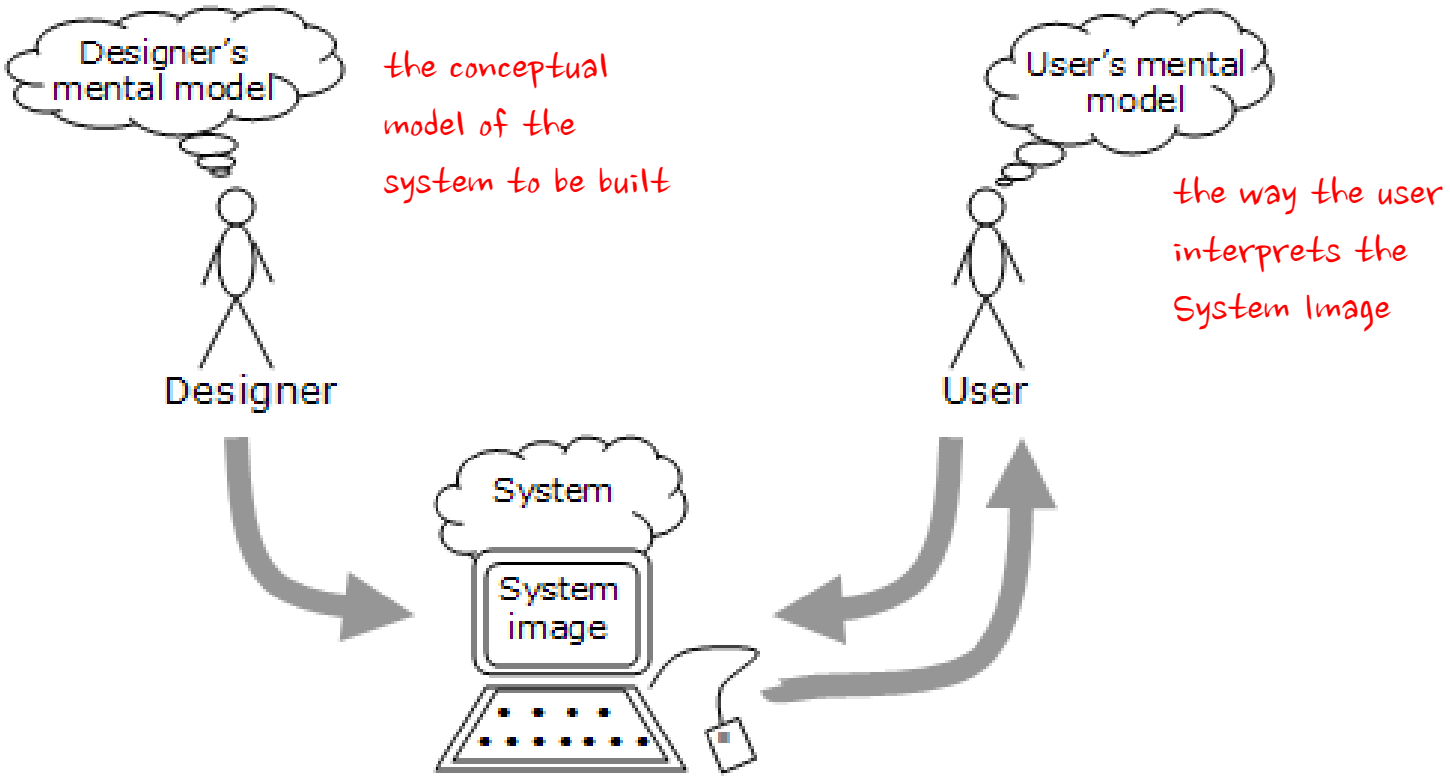
How it actually works:



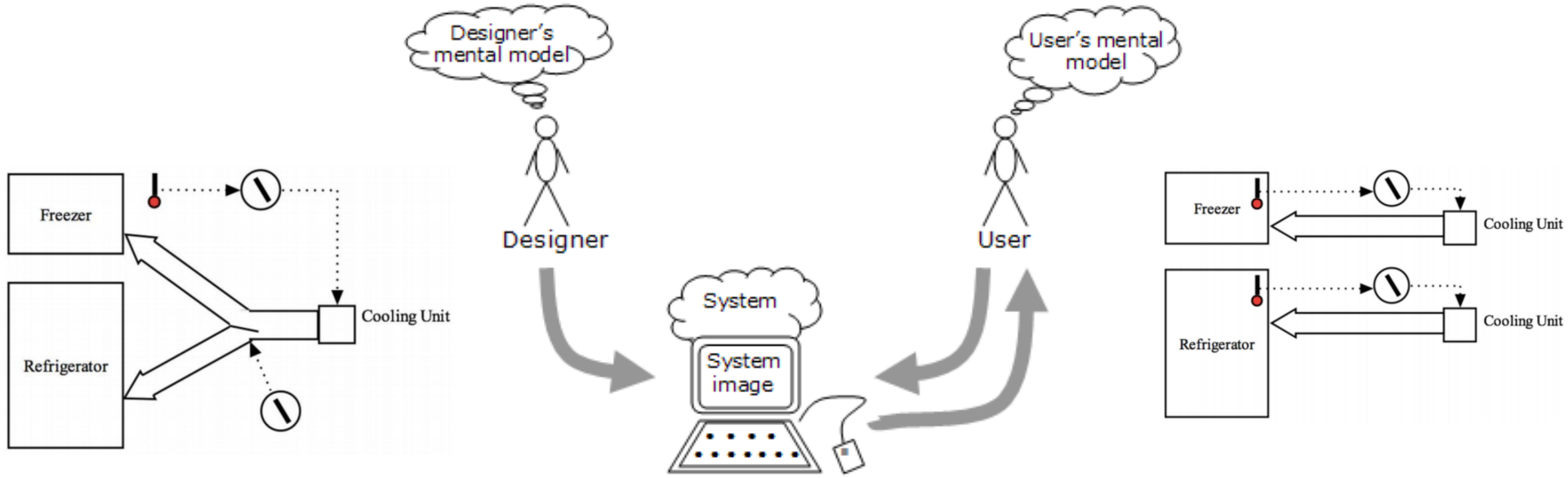
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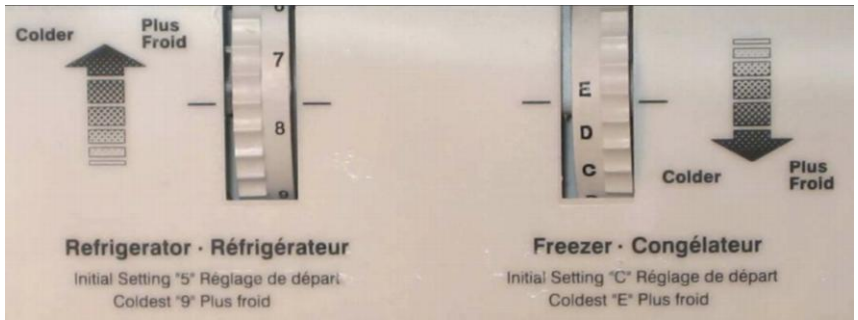


User's vs. designer's model



Mismatch between User's and Designer's model

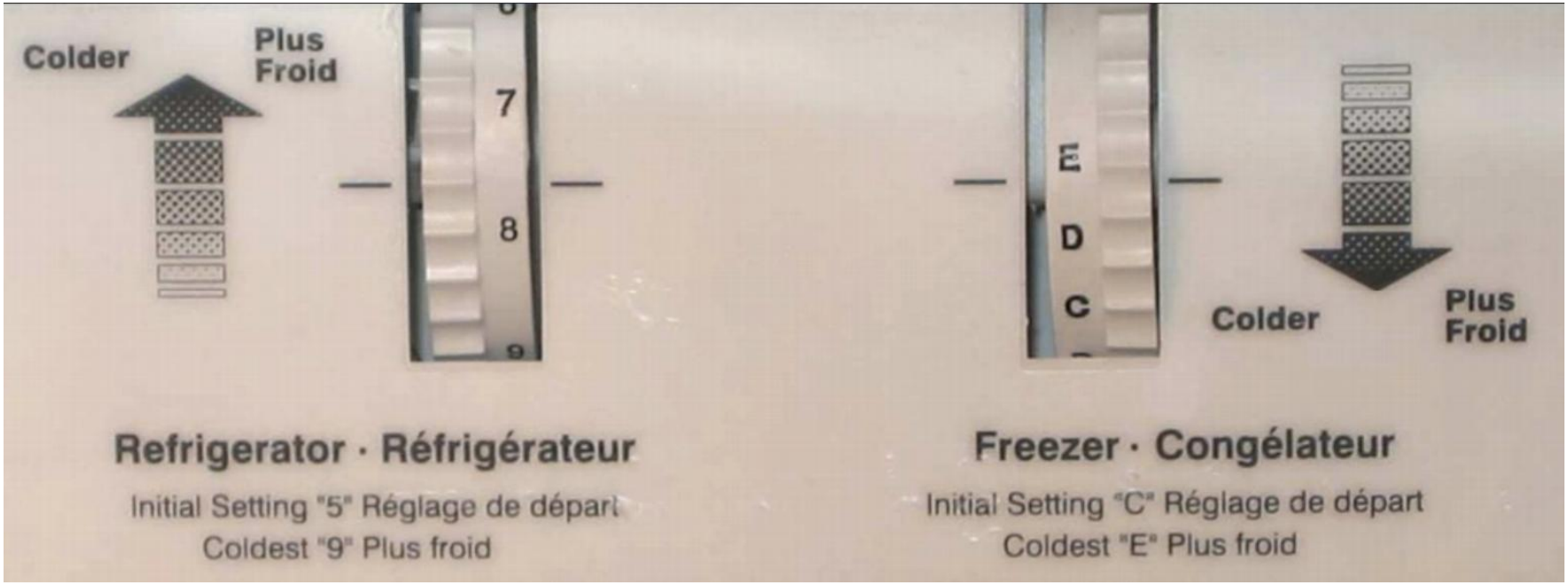




Mental model mismatches lead to slow performance, frustration, and errors

Two main kinds of errors:

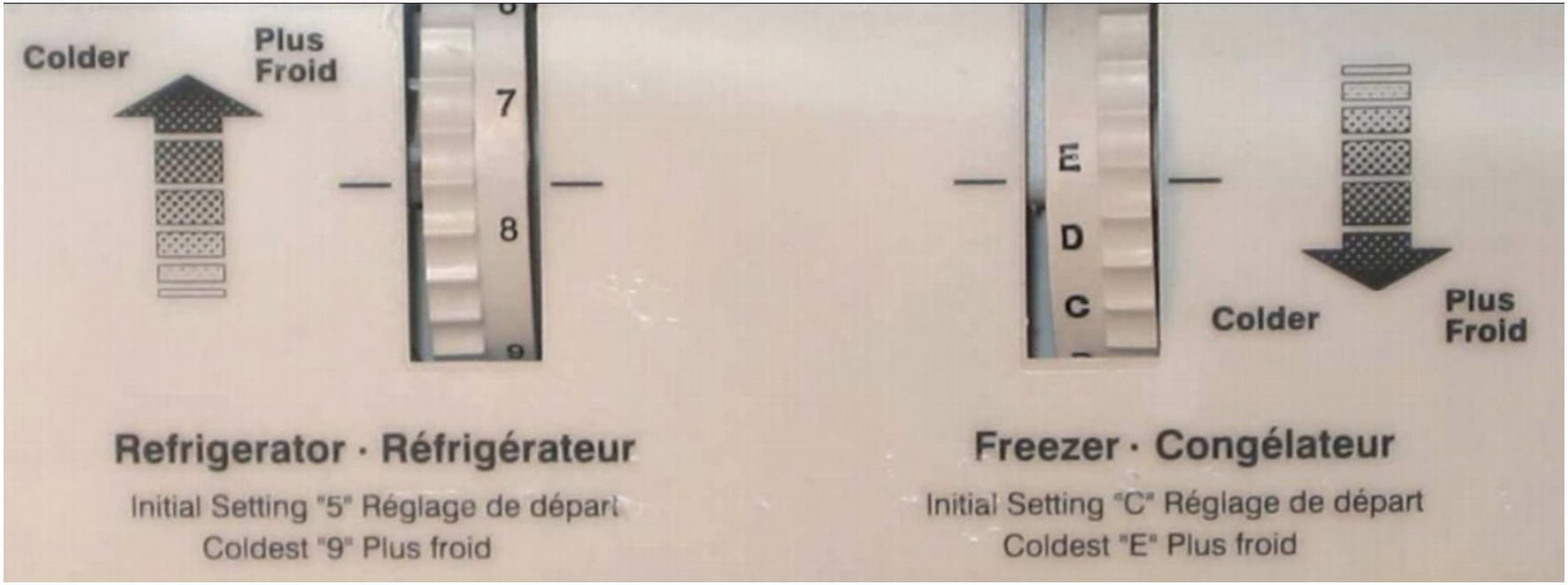
- Slips - you have correct mental model but mess up in performing the action (you're distracted by groceries, so turn the wrong dial)
- Mistakes - you perform the action you intended (turn what you think is the right dial), but have wrong mental model



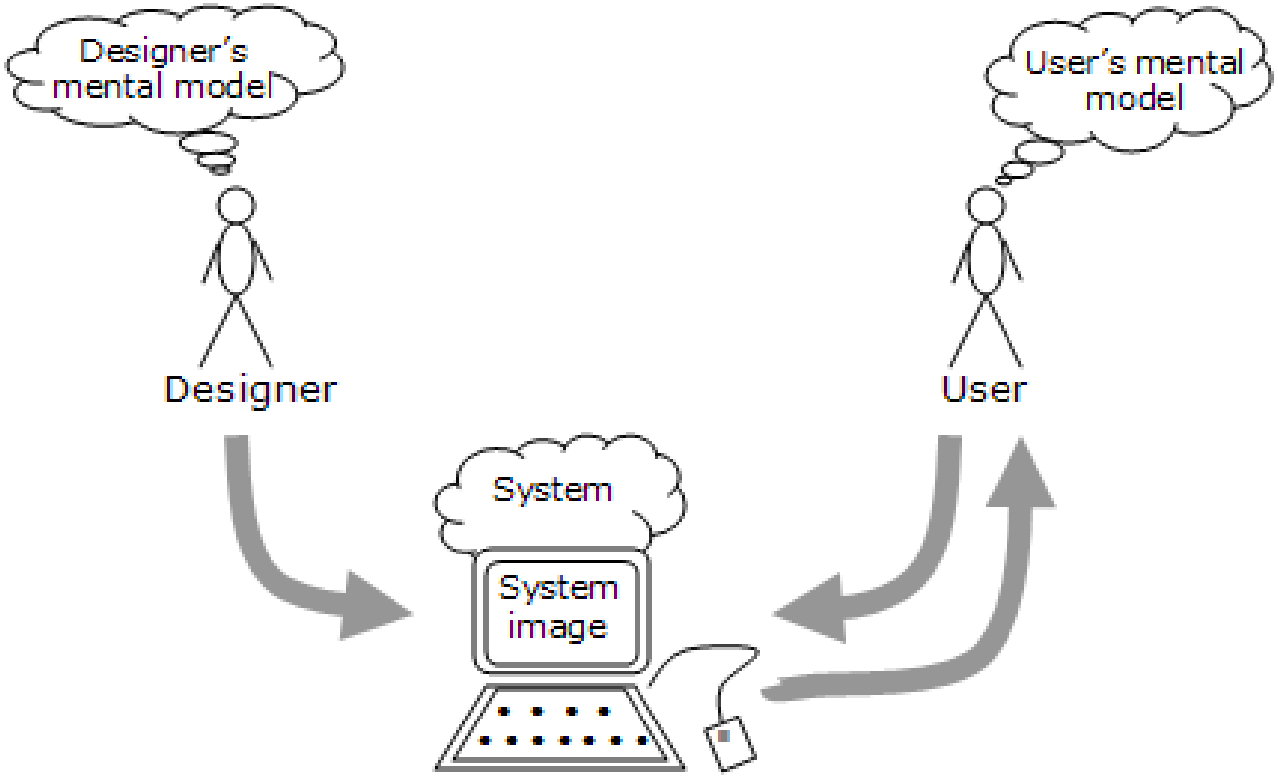
Mental model mismatches lead to slow performance, frustration, and errors

How a designer can prevent the two main kinds of errors:

- Slips - prevent by having better visual/tactile aesthetics, larger hit targets, more salient spatial layouts [easier]
- Mistakes - prevent by better conveying the designer's mental model to the user [harder]

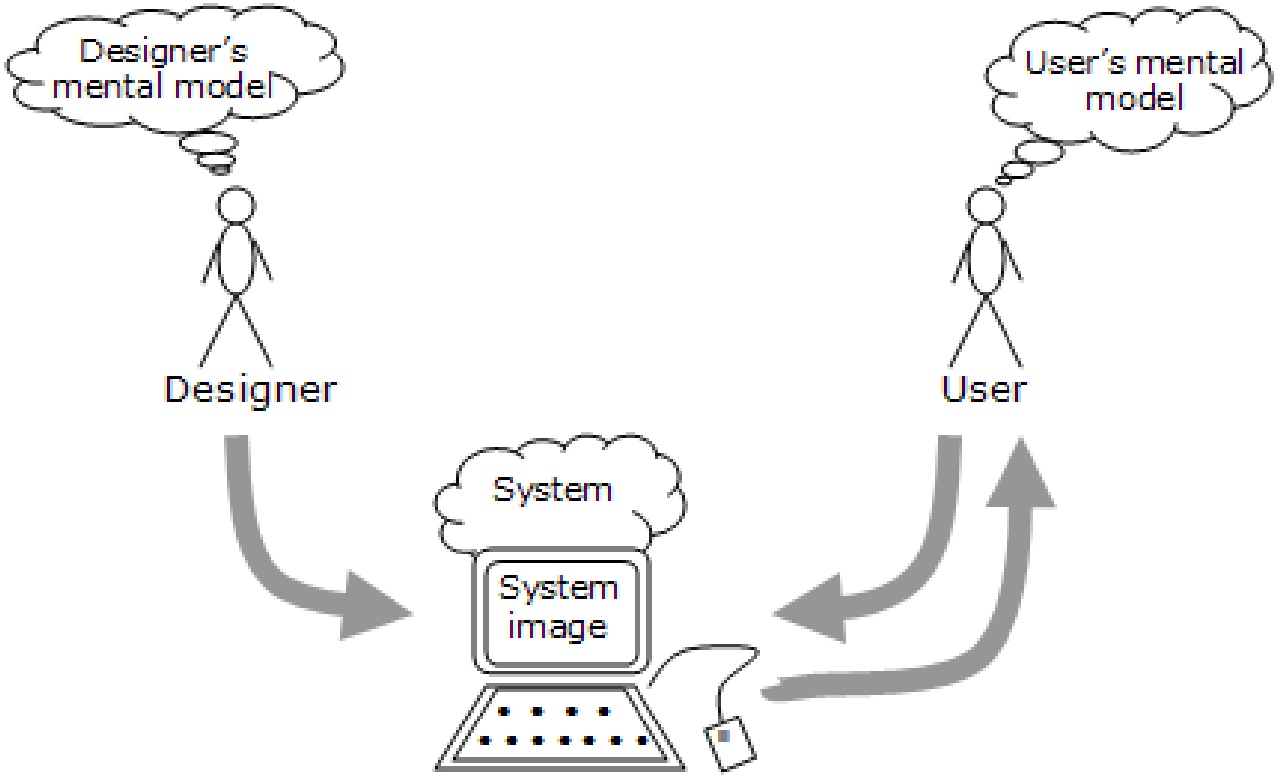


User's vs. designer's model



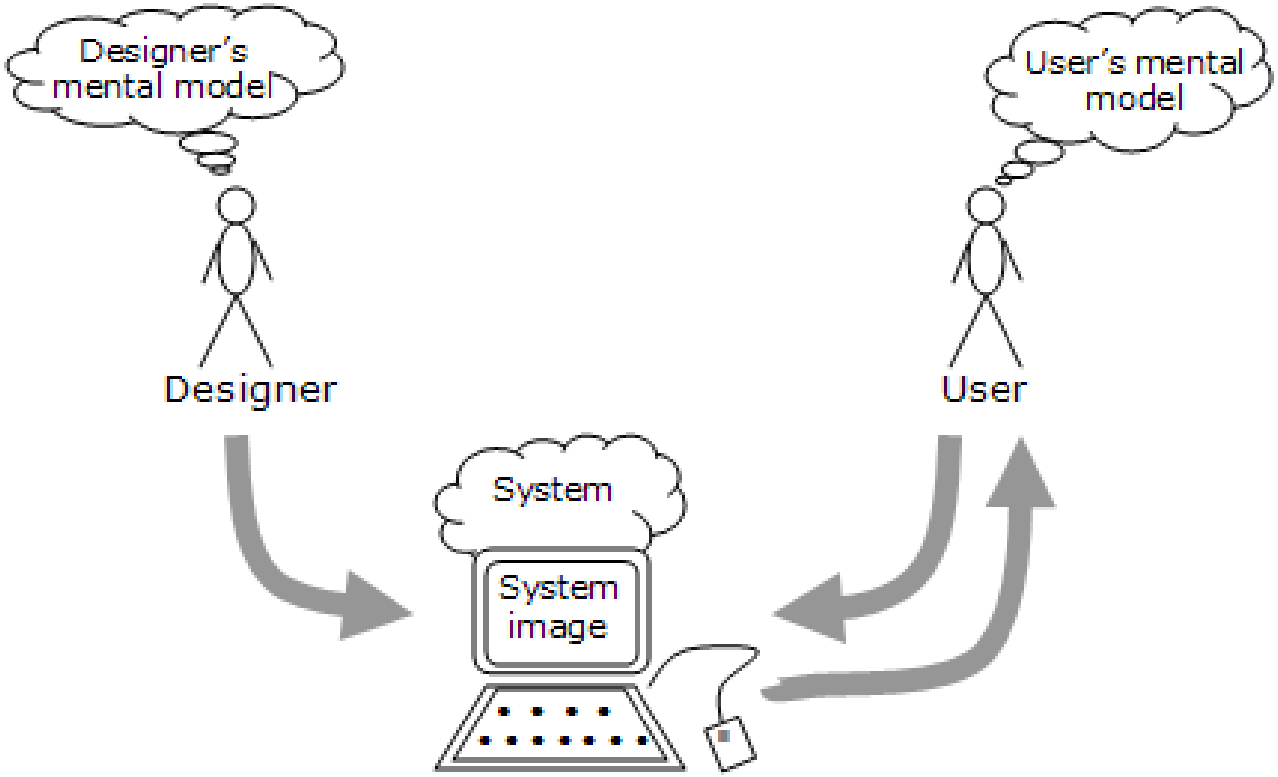
As a designer, you want to use the system's UI to accurately convey your mental model of the system to the user.

User's vs. designer's model



It is difficult because the designer is often way too familiar with their own system to empathize with what users don't know.

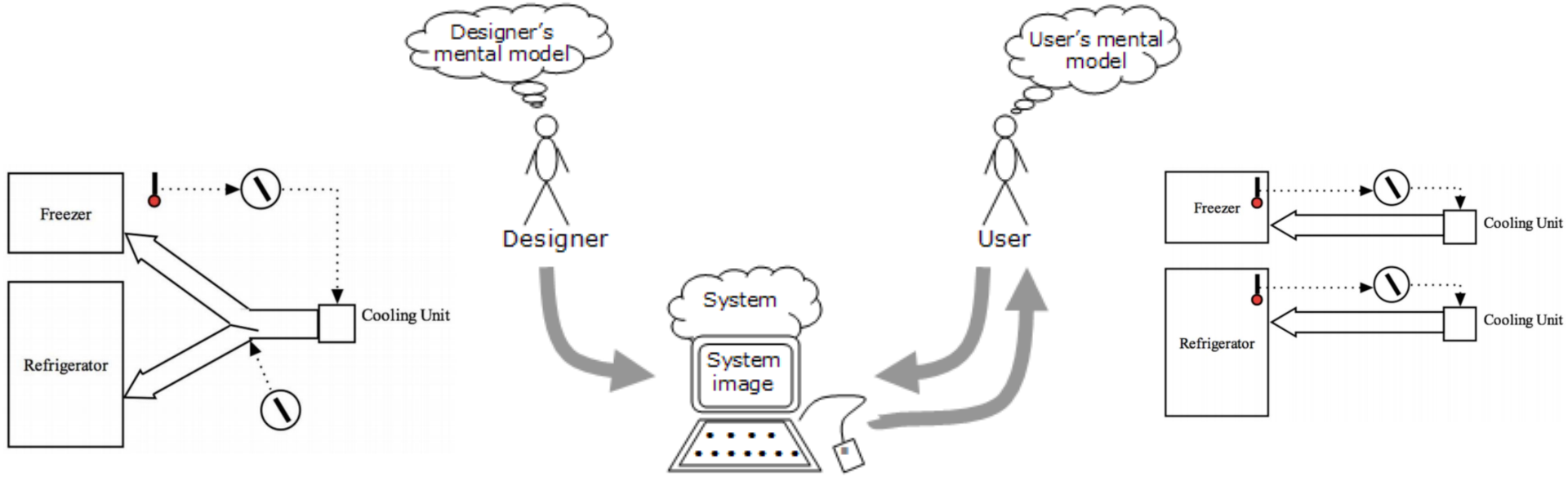
User's vs. designer's model

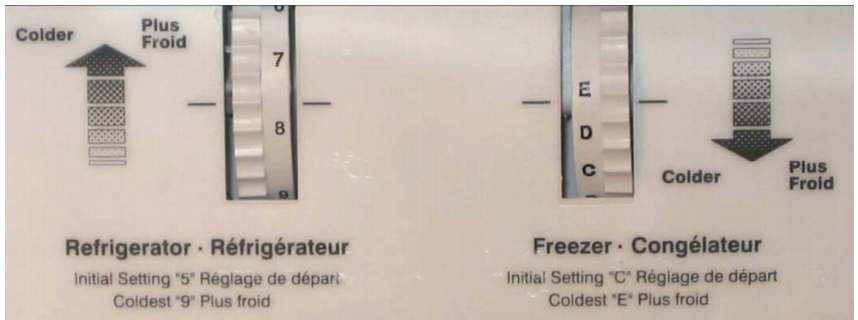


The system's UI is all that you have.

You can't directly talk to the user!

Designers of this fridge control must have thought that their mental model was the “obvious” one, which is not!

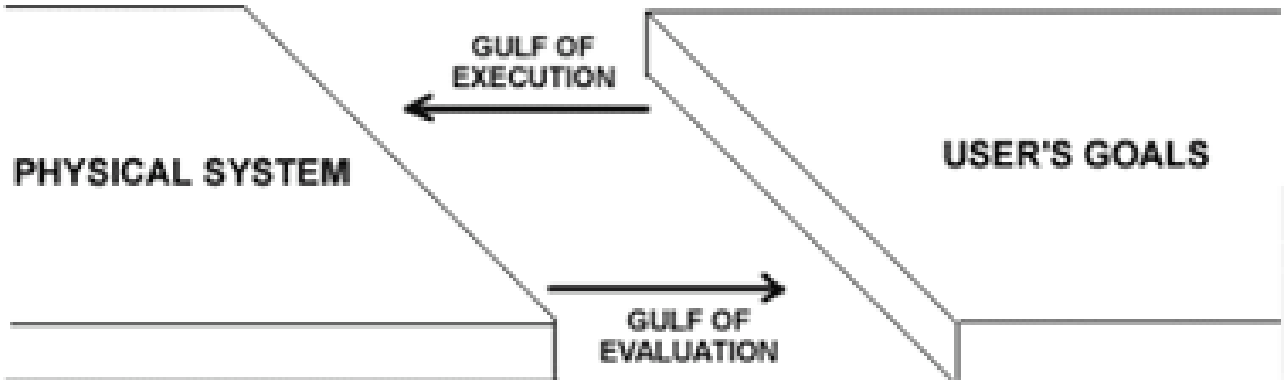




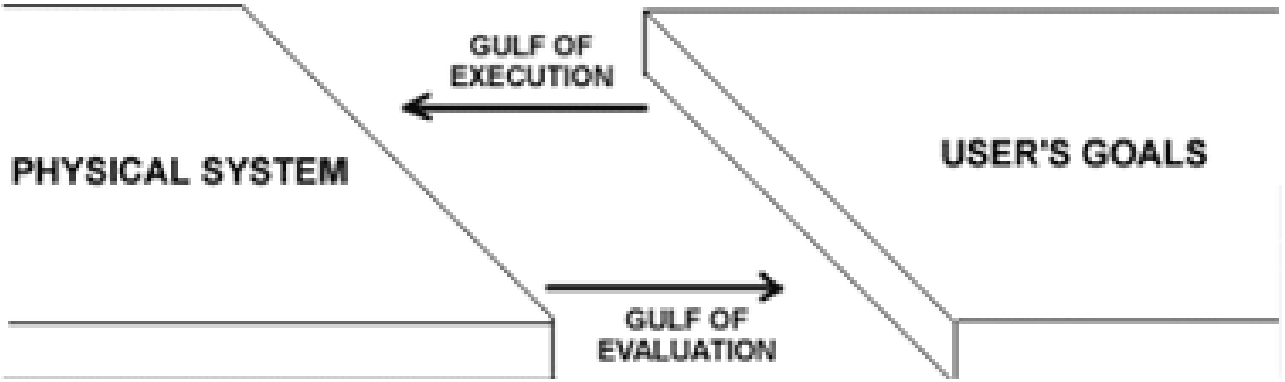
Gulfs of Execution & Evaluation

The gaps between the user and the interface, pointing to how to better design an interface so that the user can cope with it

- Gulf of execution: the distance between the user's goals and the means of achieving them through the system
- Gulf of evaluation: the amount of effort required to determine the system state



How can you close the gap between the designer's and user's mental models?



How can you close the gap between the designer's and user's mental models?

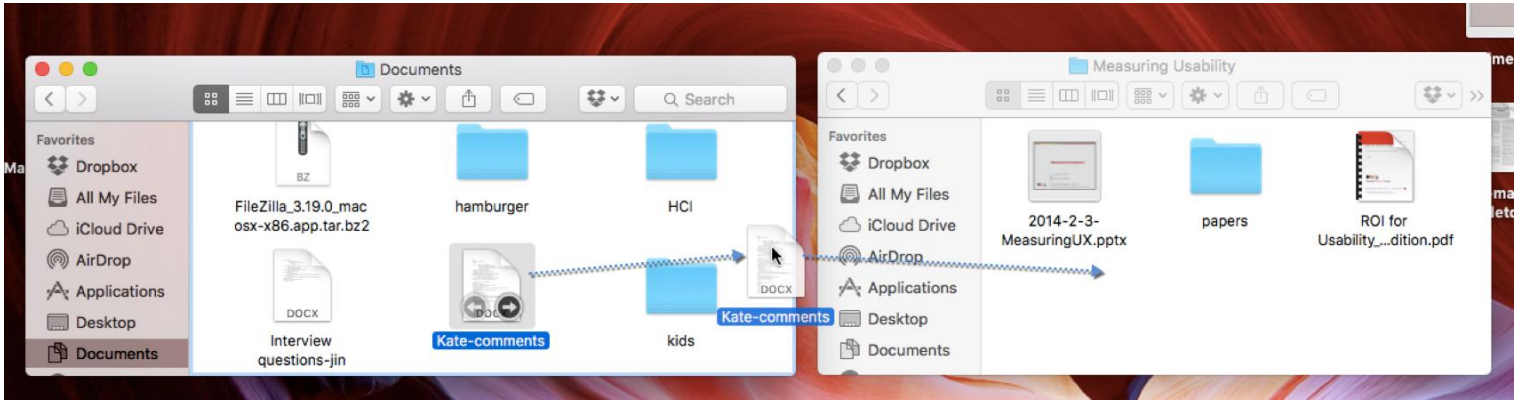
- Test on real users, and get them to convey their mental models to you
- Good design principles can more faithfully convey the designer's mental model to the user

Direct Manipulation

An interaction style in which the objects of interest in the UI are visible and can be acted upon via physical, reversible, incremental actions that receive immediate feedback.







Direct Manipulation

An interaction style in which the objects of interest in the UI are visible and can be acted upon via physical, reversible, incremental actions that receive immediate feedback.

- Immediate feedback on actions
- Continuous representations of objects
- Real world metaphors / mental models

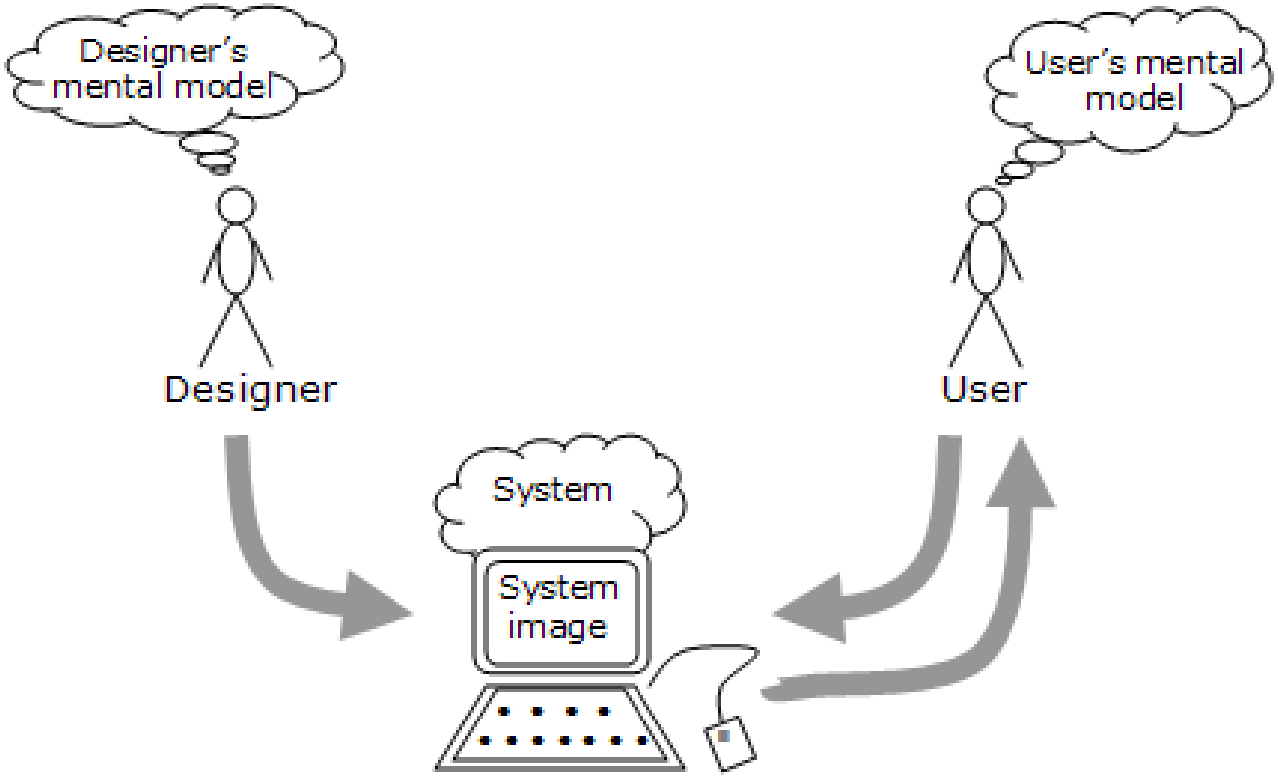
→ Direct manipulation can minimize the gap

What happens in good designs?

- Good idea of how each object works and how to control it
- Interface itself discloses how it is used

→ The art in design is to translate users cognitive capabilities and existing mental models into interfaces that work!

User's vs. designer's model



As a designer, you want to use the system's UI to accurately convey your mental model of the system to the user.

By next class

Reading

- The Psychology of Human-Computer Interaction, Ch.2